

CRM Gamification System

Roles & Administration Documentation

1. Purpose of This Document

This document defines the **roles, permissions, and administrative limitations** for the CRM + Gamification system. It is intended to guide: - Backend authorization (RBAC) - Frontend UI access control - MERN-based implementation decisions

The system is designed as a **lightweight, modern, educational CRM** inspired by tools like ClickUp, with a unique **gamification + mood-aware experience**.

2. Role Hierarchy Overview

The system supports **three roles** with clearly separated responsibilities:

1. **System Owner** – System-level authority
2. **Project Admin (Project Manager)** – Project-level authority
3. **Team Member** – Execution-level role

Role hierarchy (highest → lowest):

```
System Owner
  ↓
Project Admin
  ↓
Team Member
```

3. System Owner

3.1 Role Description

The **System Owner** owns and controls the entire system. This role is responsible for configuration, governance, and safety rather than daily task execution.

There are typically **very few system owners** (1–3 maximum).

3.2 Permissions & Access

User Management

- Create and delete users
- Assign or change system roles
- Suspend or deactivate accounts
- Force password resets

Project Management

- Create and delete projects
- Assign Project Admins to projects
- Add or remove users from any project
- Archive projects

Gamification Engine

- Define reward rules
- Configure ETH coin economy
- Set motivation score calculation logic
- Configure mood-based multipliers
- Enable or disable gamification per project
- View global leaderboards

AI & Sentiment Analysis

- Configure LLM behavior
- Set sentiment thresholds
- Manage AI usage limits
- View AI analytics and logs
- Enable or disable AI per project or user

System Configuration

- Notification types and priorities
- System-wide settings
- Audit logs and monitoring

3.3 Restrictions

- Cannot participate in tasks
 - Cannot interact in Kanban boards
 - Cannot join chats (except for debugging)
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4. Project Admin (Project Manager)

4.1 Role Description

The **Project Admin** manages people, tasks, and workflows **within a project**. This role ensures productivity and oversees gamification outcomes at the project level.

A user may be a Project Admin in one project and a Team Member in another.

4.2 Permissions & Access

Dashboard

- Full project dashboard access
- View team-wide task statistics
- View aggregated motivation and mood trends

Task & Board Management

- Create, update, and delete tasks
- Assign tasks to team members
- Set task type, priority, and deadlines
- Create and manage Kanban boards
- Move tasks across board columns
- Override task completion status

Inbox & Communication

- View project-level notifications
- Send direct messages to any project member
- Receive task and reward alerts

Team Management

- Add or remove users from the project
- Change roles within the project (Admin / Member)
- View individual performance summaries

Gamification Controls (Limited)

- Cannot change core reward algorithms
- Can manually grant rewards (with justification)
- Can flag suspicious or abusive behavior
- Can disable rewards for specific tasks
- View project-level leaderboard

AI Assistant

- Request task summaries
 - Get productivity insights
 - Ask for team-level suggestions
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4.3 Restrictions

- Cannot create or delete users globally
 - Cannot modify system-wide gamification rules
 - Cannot access system logs
 - Cannot view private chats between other users
-

5. Team Member

5.1 Role Description

The **Team Member** is focused on task execution, collaboration, and personal productivity. This role interacts most with gamification features.

5.2 Permissions & Access

Dashboard

- View personal dashboard
- See recent tasks
- View assigned tasks
- Task overview (to-do / done)
- Personal inbox preview

Mood & Motivation

- Required to submit mood based on time of day:
 - Morning (before 12 PM)
 - Evening (before 6 PM)
 - Night (after 6 PM)
- Can manually update mood
- View personal mood history
- Motivation score visible only to the user

Tasks

- View assigned tasks
- Read-only view of boards
- Update task status
- Add comments to tasks

Communication

- One-to-one chats only
- Add contacts from same project
- Messaging restricted to project members

Inbox

- Receive notifications for:
 - Task assignments
 - Rewards
 - Messages
- System notifications cannot be disabled

Gamification

- Earn ETH coins and motivation score
- View personal reward history
- View project leaderboard (read-only)

AI Assistant

- Ask for help related to own tasks
- Receive productivity and focus tips
- AI access restricted to personal data

5.3 Restrictions

- Cannot create or delete projects
 - Cannot add or remove users
 - Cannot create or manage boards
 - Cannot reassign tasks
 - Cannot view other users' moods or motivation scores
-

6. Permissions Summary Table

Feature	System Owner	Project Admin	Team Member
User Creation	Yes	No	No
Role Assignment	Yes	Project-only	No
Project Creation	Yes	No	No
Task Management	No	Yes	Assigned-only
Feature	System Owner	Project Admin	Team Member
Board Creation	No	Yes	No
Gamification Rules	Yes	No	No
Manual Rewards	No	Yes	No
Mood Submission	No	No	Yes
AI Access	Full	Limited	Personal
System Logs	Yes	No	No

7. MERN Authorization Model (Recommended)

To support flexible role management, the following structure is recommended:

```
User
├─ systemRole: SUPER_ADMIN | USER
└─ projectRoles:
    ├─ projectId
    └─ role: ADMIN | MEMBER
```

Benefits

- Clean separation of concerns
- Simple RBAC middleware
- Easy future scalability
- Clear frontend access control

8. Conclusion

This role-based design ensures: - Clear administrative boundaries - Minimal feature creep - Strong alignment with project scope - Smooth MERN implementation

This document should be treated as the **authoritative reference** during backend and frontend development.
